

● MEDIUM

● ADVANCED MATH

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# Nonlinear equations in one variable and systems of equations in two variables

01

10 questions · ~15 min

Q1

ADVANCED MATH

If  $4\sqrt{2x} = 16$ , what is the value of  $6x$ ?

- A. 24
- B. 48
- C. 72
- D. 96

Q2

ADVANCED MATH

$$5x^2 - 37x - 24 = 0$$

What is the positive solution to the given equation?

- A.  $\frac{3}{5}$
- B. 3
- C. 8
- D. 37

**Q3**

ADVANCED MATH

$$\begin{aligned}x^2 &= 6x + y \\ y &= -6x + 36\end{aligned}$$

A solution to the given system of equations is  $(x, y)$ . Which of the following is a possible value of  $xy$  ?

- A. 0
- B. 6
- C. 12
- D. 36

**Q4**

ADVANCED MATH

If  $\frac{42}{x} = 7x$ , what is the value of  $7x^2$ ?

- A. 6
- B. 7
- C. 42
- D. 294

Q5

ADVANCED MATH

$$14j + 5k = m$$

The given equation relates the numbers  $j$ ,  $k$ , and  $m$ . Which equation correctly expresses  $k$  in terms of  $j$  and  $m$ ?

- A.  $k = \frac{m - 14j}{5}$
- B.  $k = \frac{1}{5}m - 14j$
- C.  $k = \frac{14j - m}{5}$
- D.  $k = 5m - 14j$

Q6

ADVANCED MATH

$$-4x^2 - 7x = -36$$

What is the positive solution to the given equation?

- A.  $\frac{7}{4}$
- B.  $\frac{9}{4}$
- C. 4
- D. 7

Q7

ADVANCED MATH

$$T = 0.01(P - 40,000)$$

In a city, the property tax  $T$ , in dollars, is calculated using the formula above, where  $P$  is the value of the property, in dollars. Which of the following expresses the value of the property in terms of the property tax?

- A.  $P = 100T - 400$
- B.  $P = 100T + 400$
- C.  $P = 100T - 40,000$
- D.  $P = 100T + 40,000$

Q8

ADVANCED MATH

$$5x + 42x - 5 = 0$$

Which of the following is a solution to the given equation?

- A.  $-\frac{5}{2}$
- B.  $-\frac{5}{4}$
- C.  $-\frac{4}{5}$
- D.  $-\frac{2}{5}$

$$p = 20 + \frac{16}{n}$$

The given equation relates the numbers  $p$  and  $n$ , where  $n$  is not equal to 0 and  $p > 20$ . Which equation correctly expresses  $n$  in terms of  $p$ ?

A.  $n = \frac{p - 20}{16}$

B.  $n = \frac{p}{16} + 20$

C.  $n = \frac{p}{16} - 20$

D.  $n = \frac{16}{p - 20}$

$$y = x^2 - 1$$

$$y = 3$$

When the equations above are graphed in the  $xy$ -plane, what are the coordinates  $(x, y)$  of the points of intersection of the two graphs?

A. and  $(2, 3)$

B. and  $(-2, 3)$

C. and  $(2, 4)$

D. and  $(-2, 4)$

E. and  $(3, 8)$

F. and  $(-3, 8)$

G. and  $(\sqrt{2}, 3)$

H. and  $(-\sqrt{2}, 3)$